

Notes and Solutions for Semiparametric Theory and Missing Data

A companion to Tsiatis (2006)

Lu Mao

August 20, 2026

Table of contents

Home	6
How to use the book	6
Preface	7
Purpose	7
Conventions	7
Scope and attribution	7
1 Notation	8
I I. Semiparametric foundations	9
2 Introduction to semiparametric models	10
2.1 Learning objectives	10
2.2 Prerequisites and notation	10
2.3 Expository notes	10
2.4 Expanded derivations	10
2.5 Worked examples	10
2.6 Additional exercises	10
2.7 Chapter summary	10
3 Hilbert space for random vectors	11
3.1 Learning objectives	11
3.2 Prerequisites and notation	11
3.3 Expository notes	11
3.4 Expanded derivations	11
3.5 Worked examples	11
3.6 Solutions to exercises	11
3.7 Additional exercises	11
3.8 Chapter summary	11
4 The geometry of influence functions	12
4.1 Learning objectives	12
4.2 Prerequisites and notation	12
4.3 Expository notes	12
4.4 Expanded derivations	12
4.5 Worked examples	12
4.6 Solutions to exercises	12
4.7 Additional exercises	12
4.8 Chapter summary	12

5	Semiparametric models	13
5.1	Learning objectives	13
5.2	Prerequisites and notation	13
5.3	Expository notes	13
5.4	Expanded derivations	13
5.5	Worked examples	13
5.6	Solutions to exercises	13
5.7	Additional exercises	13
5.8	Chapter summary	13
6	Other examples of semiparametric models	14
6.1	Learning objectives	14
6.2	Prerequisites and notation	14
6.3	Expository notes	14
6.4	Expanded derivations	14
6.5	Worked examples	14
6.6	Solutions to exercises	14
6.7	Additional exercises	14
6.8	Chapter summary	14
II	II. Missing and coarsened data	15
7	Models and methods for missing data	16
7.1	Learning objectives	16
7.2	Prerequisites and notation	16
7.3	Expository notes	16
7.4	Expanded derivations	16
7.5	Worked examples	16
7.6	Solutions to exercises	16
7.7	Additional exercises	16
7.8	Chapter summary	16
8	Missing and coarsening at random for semiparametric models	17
8.1	Learning objectives	17
8.2	Prerequisites and notation	17
8.3	Expository notes	17
8.4	Expanded derivations	17
8.5	Worked examples	17
8.6	Solutions to exercises	17
8.7	Additional exercises	17
8.8	Chapter summary	17
9	Inverse probability weighted complete-case estimators	18
9.1	Learning objectives	18
9.2	Prerequisites and notation	18
9.3	Expository notes	18
9.4	Expanded derivations	18

9.5	Worked examples	18
9.6	Solutions to exercises	18
9.7	Additional exercises	18
9.8	Chapter summary	18
10	Semiparametric estimators with monotone coarsening	19
10.1	Learning objectives	19
10.2	Prerequisites and notation	19
10.3	Expository notes	19
10.4	Expanded derivations	19
10.5	Worked examples	19
10.6	Solutions to exercises	19
10.7	Additional exercises	19
10.8	Chapter summary	19
III	III. Efficiency and robustness	20
11	Improving efficiency and double robustness with coarsened data	21
11.1	Learning objectives	21
11.2	Prerequisites and notation	21
11.3	Expository notes	21
11.4	Expanded derivations	21
11.5	Worked examples	21
11.6	Solutions to exercises	21
11.7	Additional exercises	21
11.8	Chapter summary	21
12	Locally efficient estimators for coarsened-data semiparametric models	22
12.1	Learning objectives	22
12.2	Prerequisites and notation	22
12.3	Expository notes	22
12.4	Expanded derivations	22
12.5	Worked examples	22
12.6	Solutions to exercises	22
12.7	Additional exercises	22
12.8	Chapter summary	22
13	Approximate methods for gaining efficiency	23
13.1	Learning objectives	23
13.2	Prerequisites and notation	23
13.3	Expository notes	23
13.4	Expanded derivations	23
13.5	Worked examples	23
13.6	Solutions to exercises	23
13.7	Additional exercises	23
13.8	Chapter summary	23

IV	IV. Causal inference and imputation	24
14	Doubly robust estimation of average causal treatment effects	25
14.1	Learning objectives	25
14.2	Prerequisites and notation	25
14.3	Expository notes	25
14.4	Expanded derivations	25
14.5	Worked examples	25
14.6	Solutions to exercises	25
14.7	Additional exercises	25
14.8	Chapter summary	25
15	Multiple imputation from a frequentist perspective	26
15.1	Learning objectives	26
15.2	Prerequisites and notation	26
15.3	Expository notes	26
15.4	Expanded derivations	26
15.5	Worked examples	26
15.6	Additional exercises	26
15.7	Chapter summary	26
	References	27
	Appendices	28
A	Linear algebra and projections	28
A.1	Matrix differentiation	28
A.2	Block-matrix inversion	28
A.3	Orthogonal projections	28
B	Probability and asymptotic tools	29
B.1	Conditional expectation	29
B.2	Modes of convergence	29
B.3	Central limit arguments	29
C	R code and reproducibility	30
C.1	Session information	30

Home

This book develops explanatory notes, derivations, worked examples, and original solutions for exercises accompanying *Semiparametric Theory and Missing Data* by Anastasios A. Tsiatis.

The objective is to make the geometry of semiparametric models and the theory of missing and coarsened data easier to learn and apply. The presentation follows the organization of the source book while remaining independently written.

How to use the book

Each chapter contains the following components:

- Learning objectives and prerequisites
- A notation guide
- Expository notes and geometric intuition
- Expanded derivations
- Worked examples
- Solutions to selected exercises
- Additional exercises and computational illustrations
- A short chapter summary

Source text

Readers should consult Tsiatis (2006) for the original statements, examples, and exercises. Exercise numbers in this companion refer to that text.

Preface

Purpose

These notes are intended for graduate students and researchers in biostatistics and statistics who want a detailed path through semiparametric theory, influence functions, missing-data methods, and causal inference.

Conventions

- Random variables are written in uppercase and realizations in lowercase
- Vectors are treated as column vectors unless stated otherwise
- Expectations and inner products are taken under the true data-generating distribution unless annotated
- A solution labeled “Exercise $j.k$ ” corresponds to that numbered exercise in Tsiatis (2006)

Scope and attribution

This is a companion volume, not a reproduction of the source. Explanations, derivations, examples, and solutions should be written independently. Page and exercise references identify where the corresponding topic appears in the source.

1 Notation

Symbol	Meaning
Z	Full observed random vector for one independent unit
$Z_1, \dots, Z_n$	Independent and identically distributed observations
β	Finite-dimensional parameter of interest
η	Nuisance parameter, often infinite-dimensional
S_β	Score for β
Λ	Tangent space
$\Lambda^\perp$	Orthogonal complement of a tangent space
φ	Influence function
φ_{eff}	Efficient influence function
R	Indicator that the full data are observed
π	Observation or response probability

Add chapter-specific notation to this table as the manuscript develops.

Part I

I. Semiparametric foundations

2 Introduction to semiparametric models

2.1 Learning objectives

- Distinguish parametric, nonparametric, and semiparametric models
- Identify parameters of interest and nuisance parameters
- Recognize restricted moment and proportional hazards models as semiparametric

2.2 Prerequisites and notation

2.3 Expository notes

2.4 Expanded derivations

2.5 Worked examples

2.6 Additional exercises

2.7 Chapter summary

3 Hilbert space for random vectors

3.1 Learning objectives

- Work with mean-zero, square-integrable random functions
- Interpret orthogonality through covariance
- Use the projection theorem in statistical problems

3.2 Prerequisites and notation

3.3 Expository notes

3.4 Expanded derivations

3.5 Worked examples

3.6 Solutions to exercises

3.6.1 Exercise 2.1

Solution. Key idea.

Derivation.

Result and interpretation.

3.7 Additional exercises

3.8 Chapter summary

4 The geometry of influence functions

4.1 Learning objectives

- Define regular asymptotically linear estimators
- Derive influence functions for M-estimators
- Characterize the efficient influence function geometrically

4.2 Prerequisites and notation

4.3 Expository notes

4.4 Expanded derivations

4.5 Worked examples

4.6 Solutions to exercises

4.7 Additional exercises

4.8 Chapter summary

5 Semiparametric models

5.1 Learning objectives

- Construct parametric submodels
- Define nuisance tangent spaces
- Derive influence functions and efficiency bounds in semiparametric models

5.2 Prerequisites and notation

5.3 Expository notes

5.4 Expanded derivations

5.5 Worked examples

5.6 Solutions to exercises

5.7 Additional exercises

5.8 Chapter summary

6 Other examples of semiparametric models

6.1 Learning objectives

- Analyze location-shift regression models
- Connect proportional hazards estimation to tangent-space geometry
- Derive efficient estimators for means and treatment contrasts

6.2 Prerequisites and notation

6.3 Expository notes

6.4 Expanded derivations

6.5 Worked examples

6.6 Solutions to exercises

6.7 Additional exercises

6.8 Chapter summary

Part II

II. Missing and coarsened data

7 Models and methods for missing data

7.1 Learning objectives

- Distinguish MCAR, MAR, and nonignorable missingness
- Compare likelihood, imputation, and inverse-probability methods
- Explain the basis of doubly robust estimation

7.2 Prerequisites and notation

7.3 Expository notes

7.4 Expanded derivations

7.5 Worked examples

7.6 Solutions to exercises

7.7 Additional exercises

7.8 Chapter summary

8 Missing and coarsening at random for semiparametric models

8.1 Learning objectives

- Formalize missing and coarsened data
- State coarsening-at-random assumptions
- Relate full-data and observed-data models

8.2 Prerequisites and notation

8.3 Expository notes

8.4 Expanded derivations

8.5 Worked examples

8.6 Solutions to exercises

8.7 Additional exercises

8.8 Chapter summary

9 Inverse probability weighted complete-case estimators

9.1 Learning objectives

- Construct IPW complete-case estimating equations
- Establish consistency and asymptotic normality
- Estimate asymptotic variance when response probabilities are estimated

9.2 Prerequisites and notation

9.3 Expository notes

9.4 Expanded derivations

9.5 Worked examples

9.6 Solutions to exercises

9.7 Additional exercises

9.8 Chapter summary

10 Semiparametric estimators with monotone coarsening

10.1 Learning objectives

- Represent monotone coarsening processes
- Derive semiparametric estimators under monotone missingness
- Connect censoring to monotone coarsening

10.2 Prerequisites and notation

10.3 Expository notes

10.4 Expanded derivations

10.5 Worked examples

10.6 Solutions to exercises

10.7 Additional exercises

10.8 Chapter summary

Part III

III. Efficiency and robustness

11 Improving efficiency and double robustness with coarsened data

11.1 Learning objectives

- Project influence functions onto augmentation spaces
- Construct efficient augmented IPW estimators
- Explain double robustness under different coarsening patterns

11.2 Prerequisites and notation

11.3 Expository notes

11.4 Expanded derivations

11.5 Worked examples

11.6 Solutions to exercises

11.7 Additional exercises

11.8 Chapter summary

12 Locally efficient estimators for coarsened-data semiparametric models

12.1 Learning objectives

- Derive observed-data efficient scores
- Compare likelihood and augmented-IPW representations
- Construct locally efficient estimators

12.2 Prerequisites and notation

12.3 Expository notes

12.4 Expanded derivations

12.5 Worked examples

12.6 Solutions to exercises

12.7 Additional exercises

12.8 Chapter summary

13 Approximate methods for gaining efficiency

13.1 Learning objectives

- Define restricted classes of augmented estimators
- Derive optimal estimators within a restricted class
- Balance efficiency gains against computational feasibility

13.2 Prerequisites and notation

13.3 Expository notes

13.4 Expanded derivations

13.5 Worked examples

13.6 Solutions to exercises

13.7 Additional exercises

13.8 Chapter summary

Part IV

IV. Causal inference and imputation

14 Doubly robust estimation of average causal treatment effects

14.1 Learning objectives

- State identification assumptions for point-treatment effects
- Express causal estimation as a coarsened-data problem
- Derive the augmented inverse-probability estimator for the ATE

14.2 Prerequisites and notation

14.3 Expository notes

14.4 Expanded derivations

14.5 Worked examples

14.6 Solutions to exercises

14.7 Additional exercises

14.8 Chapter summary

15 Multiple imputation from a frequentist perspective

15.1 Learning objectives

- Compare full-data and observed-data information
- Derive the asymptotic distribution of multiple-imputation estimators
- Explain proper imputation and Rubin's variance estimator

15.2 Prerequisites and notation

15.3 Expository notes

15.4 Expanded derivations

15.5 Worked examples

15.6 Additional exercises

15.7 Chapter summary

References

Tsiatis, Anastasios A. (2006). Semiparametric theory and missing data. Springer. <https://doi.org/10.1007/0-387-37345-4>.

A Linear algebra and projections

This appendix will collect matrix identities, projection formulas, and block-matrix calculations used throughout the book.

A.1 Matrix differentiation

A.2 Block-matrix inversion

A.3 Orthogonal projections

B Probability and asymptotic tools

This appendix will collect conditional-expectation identities, convergence results, empirical-process notation, and asymptotic linearity arguments.

B.1 Conditional expectation

B.2 Modes of convergence

B.3 Central limit arguments

C R code and reproducibility

This appendix contains reusable R functions and reproducibility conventions for computational examples.

C.1 Session information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 10 x64 (build 19045)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8
```

```
time zone: America/Chicago
tzcode source: internal
```

```
attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
loaded via a namespace (and not attached):
[1] compiler_4.5.3    fastmap_1.2.0      cli_3.6.5          htmltools_0.5.9
[5] tools_4.5.3       otel_0.2.0         rstudioapi_0.18.0  yaml_2.3.12
[9] tinytex_0.58      rmarkdown_2.30     knitr_1.51         jsonlite_2.0.0
[13] xfun_0.56         digest_0.6.39      rlang_1.1.7        evaluate_1.0.5
```